THE BOUNDARY MAP OF A SQUARE ZERO IDEAL EXTENSION OF SHEAVES OF RINGS IS A DERIVATION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the displayed computation of the cocycle of the product; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-boundary-derivation-over-cup-product is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Boundary map of a short exact sequence of sheaves in cohomology. Same chapter, the long exact cohomology sequence of a short exact sequence of sheaves, used at proof lines 5746-5747. Supplies the map whose derivation property is being proved.
- Description of H^1 by Cech cocycles for a covering. Same chapter, Lemma lemma-cech-h1, invoked at proof lines 5766-5767. Supplies that the first Cech cohomology group of the covering embeds in M, so that the boundary classes can be represented by cocycles.
- Notion of a derivation of a ring into a module. Algebra, Definition algebra-definition-derivation, named in the statement at line 5751. Supplies the two axioms (additivity and the Leibniz rule) that the boundary map has to satisfy.

Source reference. Original definition 00RN: sources/stacks-chow/tags.txt, source lines 1009–1009. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0B8R: sources/stacks-cohomology/pinned-original-source.tex, source lines 1486–1486. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a topological space. Let $\mathcal{O}' \to \mathcal{O}$ be a surjection of sheaves of rings whose kernel $\mathcal{I} \subset \mathcal{O}'$ has square zero. Then $M = H^1(X, \mathcal{I})$ is a $R =$*

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$H^0(X, \mathcal{O})$-module and the boundary map $\partial : R \to M$ associated to the short exact sequence

$$0 \to \mathcal{I} \to \mathcal{O}' \to \mathcal{O} \to 0$$

is a derivation (Algebra, Definition 00RN).

Proof. The map $\mathcal{O}' \to \mathcal{H}om(\mathcal{I}, \mathcal{I})$ factors through $\mathcal{O}$ as $\mathcal{I} \cdot \mathcal{I} = 0$ by assumption. Hence $\mathcal{I}$ is a sheaf of $\mathcal{O}$-modules and this defines the R-module structure on M. The boundary map is additive hence it suffices to prove the Leibniz rule. Let $f \in R$. Choose an open covering $\mathcal{U} : X = \bigcup U_i$ such that there exist $f_i \in \mathcal{O}'(U_i)$ lifting $f|_{U_i} \in \mathcal{O}(U_i)$. Observe that $f_i - f_j$ is an element of $\mathcal{I}(U_i \cap U_j)$. Then $\partial(f)$ corresponds to the Čech cohomology class of the 1-cocycle α with $\alpha_{i_0 i_1} = f_{i_0} - f_{i_1}$. (Observe that by Lemma 0B8R the first Čech cohomology group with respect to $\mathcal{U}$ is a submodule of M.) Next, let $g \in R$ be a second element and assume (after possibly refining the open covering) that $g_i \in \mathcal{O}'(U_i)$ lifts $g|_{U_i} \in \mathcal{O}(U_i)$. Then we see that $\partial(g)$ is given by the cocycle β with $\beta_{i_0 i_1} = g_{i_0} - g_{i_1}$. Since $f_i g_i \in \mathcal{O}'(U_i)$ lifts $fg|_{U_i}$ we see that $\partial(fg)$ is given by the cocycle γ with

$$\gamma_{i_0 i_1} = f_{i_0} g_{i_0} - f_{i_1} g_{i_1} = (f_{i_0} - f_{i_1}) g_{i_0} + f_{i_1}(g_{i_0} - g_{i_1}) = \alpha_{i_0 i_1} g + f \beta_{i_0 i_1}$$

by our definition of the $\mathcal{O}$-module structure on $\mathcal{I}$. This proves the Leibniz rule and the proof is complete. □